

DAY 6 WEEK 1

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PROJECT: Battery SoC Estimation & Remaining Charging-Time Prediction
Using Machine Learning

Day 6: Pre-Dataset Readiness & Mentor Review

1. Methodology V1

1. Load and catalogue data

- Load all 208 LG HG2 CSV files.
- Preserve raw data; process copies only.

2. Inspect schema and metadata

- Check columns, data types, structure.
- Review file details and missing values.

3. Audit data quality

- Find duplicates and missing cells.
- Verify exact duplicates and sparse rows.

4. Perform EDA

- Visualize distributions.
- Check correlations.
- Explore temperature and other patterns.

5. Build preprocessing pipeline

- Handle missing values.
- Encode / scale data.
- Create features (if needed).

6. Train baseline and advanced models

- Test multiple models.
- Tune hyperparameters.
- Compare performance.

7. Compare, analyse errors and select final method

- Evaluate all models.
- Analyse prediction errors.
- Choose best method.

2. Experiment Plan Updated

Receive & archive raw data

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Inspect schema and metadata

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Audit data quality

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Perform EDA

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Build preprocessing pipeline

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Train baseline and advanced models

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Compare models, analyse errors and select final method